Our PCB handles the motion control and other systems required to run a fiber laser (such as gas control, focus, modulation of the laser power supply, etc.). We have decided to split these tasks between two different controllers – the Motion Control Board(MCB) and the Laser Control Board (LCB). For the MCB we chose to go with the i.MX RT1060 (similar to what is on the Teensy 4.1) running the grblHAL firmware. For the LCB we chose to use the RP2040. Both of these controllers are connected the a USB hub chip that connected to one single USB-C port.

The prototype design has been finalized, and we had 10 PCBs manufactured to test everything. Our current issue that we face is that we are unable to initiate communication with the RT1060 chip. Specifically it seems that the MKL02 bootloader chip is not starting up as expected. Here is the testing procedure that I have followed thus far...

- Verified SNVS rail: Measured VDD_SNVS_IN = 3.286 V (good; SNVS domain alive).
- Noted PROG LED behavior: Not blinking (consistent with MKL02 not running or not driving it).
- Checked global reset: POR_B = 3.294 V steady (RT1062 not being held in reset).
- Checked ONOFF: ONOFF = 3.254 V steady (SNVS requests main power-up).
- Checked core rail: VDD_SOC_IN = 0 V (internal buck not producing 1.15 V).
- Checked buck input: DCDC_IN = 3.299 V (buck has input power).
- Checked buck enable: DCDC_PSWITCH = 0 V (buck not being enabled).
- Confirmed buck hardware works: Temporarily pulled DCDC_PSWITCH → 3V3 with 10 kΩ, saw VDD_SOC_IN ≈ 1.155 V (inductor/output path OK; issue is enable control).
- Focused on MKL02 bootloader MCU (expected to drive PSWITCH):
    - MKL02 VDD = 3.281 V (powered).
    - MKL02 RESET_b (PTA1) initially measured ≈ −0.21 V (held low/loaded).
    - With power off:
        - RESET_b → GND ≈ 0.706 MΩ (no hard short).
        - RESET_b → 3V3 ≈ 65 kΩ (weak pull-up only).
    - Added 10 kΩ pull-up to 3V3: confirmed RESET_b → 3V3 ≈ 9.11 kΩ, but RESET_b still ~0.196 V when powered (something sinking it).